Introduction to AI for Teaching and Assessment: session transcript

Dr Kelly Ilich | Using AI to Enhance Literacy and Numeracy Skills, Module 2: Algorithmic literacy | Wednesday 4 November 2026

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[0:00:30] Jane Sutherland: Sayyad.

[0:00:32] Kelly Ilich: How you going?

[0:00:34] Jane Sutherland: I'm going well. Is everything okay with you?

[0:00:36] Kelly Ilich: HUM. I think so. Hopefully the technology will all go to plan. I might just get it up on my phone as well, just in case.

[0:00:39] Jane Sutherland: Good. Mhm, mhm. Are you in the office?

[0:00:51] Kelly Ilich: Yes, I am.

[0:00:51] Jane Sutherland: You at home? Okay, well, I'm downstairs if anything major happens, so...

[0:00:53] Kelly Ilich: Yeah. Okay. Perfect. Thank you. And I didn't bring a charger, so hopefully my laptop will...

[0:01:09] Jane Sutherland: Do you want me to run one up if you just got the regular work laptop?

[0:01:11] Kelly Ilich: The input. No, I've got my iMac here.

[0:01:16] Jane Sutherland: Yeah, no, I don't have my Mac adapter. Sorry.

[0:01:19] Kelly Ilich: That's all right. I'll probably just start off without sharing the PowerPoint and then...

[0:01:28] Jane Sutherland: You do whatever you...

[0:01:29] Kelly Ilich: And started, I suppose. Yeah.

[0:01:32] Jane Sutherland: I will just be, I'll log off, but if you need anything, just give me a Teams call and I can nip up.

[0:01:32] Kelly Ilich: Yeah. HUM. Right. Thank you.

[0:01:38] Jane Sutherland: Okay. Thanks, Kelly. See you.

[0:01:40] Kelly Ilich: Saying, I...

[0:02:30] Participant: No.

[0:02:32] Participant: When was I?

[0:02:33] Participant: It is.

[0:02:33] Participant: Sips it. being.

[0:02:36] Participant: One is clear that.

[0:02:36] Participant: Yeah, yeah, yeah, yeah.

[0:02:36] Participant: Yeah, about. Maybe you are something.

[0:02:41] Participant: Is.

[0:02:43] Participant: What is it?

[0:02:47] Participant: What?

[0:02:48] Participant: You. Are you?

[0:02:51] Participant: The.

[0:02:53] Participant: Playing what?

[0:02:54] Participant: And what? It's like I didn't do one thing.

[0:02:56] Participant: If to try, I'm not.

[0:02:57] Participant: I'm gonna keep it up with him.

[0:02:58] Participant: Play.

[0:02:59] Participant: Mhm. Bing.

[0:03:01] Participant: Yeah. OK, we are. So I had that. I can under here. Get him.

[0:03:01] Participant: I.

[0:03:03] Participant: Ohh.

[0:03:03] Participant: Call.

[0:03:03] Participant: Ohh. I will be doing that now.

[0:03:07] Participant: Hey, Cortana.

[0:03:12] Participant: Dala, dala, na.

[0:03:12] Participant: I agree.

[0:03:15] Participant: You got this.

[0:03:16] Participant: Hey, Cortana, I'll see you, I'll see you, you, you.

[0:03:18] Participant: No. Well, I mean, yeah.

[0:03:22] Participant: And. What?

[0:03:27] Participant: No.

[0:03:28] Participant: Play song with Doran.

[0:03:30] Participant: Yeah.

[0:03:34] Participant: The.

[0:03:35] Participant: Uh-huh.

[0:03:36] Participant: Try to get up. No, no, no, no, no, no.

[0:03:41] Participant: Play the song.

[0:03:43] Participant: No. No, I mean nothing after after the.

[0:03:47] Participant: We have to let you know.

[0:03:53] Participant: Right. We got.

[0:04:00] Participant: Yeah. Gemma.

[0:04:04] Participant: So, yeah.

[0:04:05] Participant: Oh, I see. Where are you?

[0:04:11] Participant: HUM.

[0:04:11] Participant: I understand.

[0:04:13] Participant: Uh.

[0:04:16] Participant: Simi.

[0:04:17] Participant: I know, but I know I was focused on.

[0:04:21] Participant: Yes.

[0:04:21] Participant: HUM.

[0:04:22] Participant: Cortana.

[0:04:24] Participant: The.

[0:04:26] Participant: Ms. Mmh. What we what we started? What?

[0:04:51] Participant: You know.

[0:04:55] Kelly Ilich: Okay, hello everybody. Can you see me and hear me okay? Just let me know in the chat.

[0:05:00] Participant: Okay.

[0:05:05] Kelly Ilich: HUM.

[0:05:07] Participant: Yes, we can see you and hear you very clearly.

[0:05:10] Kelly Ilich: Fantastic. Thank you so much everybody for joining. I can see we've still got lots of people coming in. And thank you so much everyone for being here and for putting on your cameras. I can see lots of your lovely faces and for muting your microphones. Very much appreciated. Okay, so I'm going to share my screen now and we'll make a start. So just give me a second. And I'm also going to hit record as well and record this session for you all. Okay, so let's do that now.

[0:05:50] Participant: Thank you, Ellie.

[0:05:51] Kelly Ilich: I believe it's already recording. and share my screen.

[0:05:56] Participant: Yeah.

[0:05:58] Kelly Ilich: Okay, hopefully you can all see my screen there. Is that okay for everybody?

[0:06:07] Participant: Yes, I bring this glitter.

[0:06:07] Participant: HUM.

[0:06:08] Kelly Ilich: Excellent. Thank you. Alright, lovely. So, welcome everybody. Hello, my name is Kelly Ilich and I work here at the School of Education at Curtin University. So it is my absolute pleasure to be delivering this session on algorithmic literacy to you all as part of the Enhancing Literacy and Numeracy Skills using AI. programme that you're all participating in. Before we do start, I would like to acknowledge the traditional custodians of the land on which I stand, that is the Whadjuk people of the Noongar Nation, and pay my respects to their elders past, present and emerging. Alright. So I thought I'd just start off by introducing myself. So as I said, I work here in the School of Education, but my background is in high school teaching. I was a secondary HASS teacher. So HASS in Western Australia stands for Humanities and Social Sciences. So, you know, it include things like politics and law, history, economics, geography. And I've mainly taught in Western

Australia since 2006. However, about four years ago, I came over to the tertiary sector and I've really been enjoying the experience of...

[0:07:29] Participant: But

[0:07:33] Participant: Yeah.

[0:07:34] Participant: But so's hot.

[0:07:35] Participant: HUM.

[0:07:35] Kelly Ilich: teaching all of our pre-service teachers just like you all do. It's a very, very rewarding career, one that I'm very happy to be doing. So if you've just joined and you've got your mics on, would you mind just muting yourself for me because we've got a little bit of background noise coming through. I really appreciate that. Thank you so much.

[0:07:37] Participant: What am I doing? Play. Other. Oh, yeah. I am.

[0:07:56] Kelly Ilich: So I've just recently finished my... Oh, sorry, we've still got some people who have their mics on, so if you can just turn them off, please.

[0:07:56] Participant: They pay. I know they are busy. Fine. The.

[0:08:12] Kelly Ilich: Let's see if I can do it from my end. There we go. Let's see if that will help. All right. Yeah, so if we make sure our mics are off, that really helps. Unless of course you've got a question, which at any point in time, please feel free to use the hands up feature and you can put your mic on then to ask a question.

[0:08:37] Participant: But what?

[0:08:37] Kelly Ilich: All right.

[0:08:38] Participant: Okay.

[0:08:40] Kelly Ilich: Oh, we still got some marks on. And we still got people arriving, OK? All right. So yes, I've just recently completed my PhD and the topic of that was looking at teachers capability to provide algorithmic literacy education in K to 12 schools in Australia. So that involved having 245 teachers complete an online survey and I also interviewed 17 teachers from across different areas of Australia, primary and secondary.

[0:09:18] Participant: Yes, Hardy.

[0:09:20] Kelly Ilich: Oh, sorry, got some microphones on again. If we can please mute those, that would be great.

[0:09:22] Participant: HUM. Play play music.

[0:09:27] Kelly Ilich: All right. And my research now involves extending my PhD studies to a cross-national study. So I've got a grant that I'm currently completing where we are comparing the provision of algorithmic literacy education and teachers capability to provide it in England, Germany, Italy and Finland. So that is a pilot project and then hopefully we'll get a larger grant in the future to be able to extend that further and have more countries participate. I'm also working with the AI and Education group at Oxford University. We're looking there at how pre-service teacher education or initial teacher education needs to change to support human flourishing in an AI era. So that's been a really interesting one to participate in as well. Okay. So what we are trying to achieve by the end of this session today, so it's a long session, we've got three hours, but I am going to, we'll have a few breaks in between so you can get up and stretch your legs and get a drink and all that kind of thing.

So the first thing that we're going to look at is 3 dimensions of algorithmic literacy, and you would have already had a bit of an introduction to that. by going through the materials on the website. We'll look at how algorithmic systems, including AI, produces outputs and what they are optimised for. We'll evaluate whose interests a system serves, how it shapes what you see, know, and decide. and will apply strategies to shape AI output, such as asking it to challenge you in a critical manner. Okay, so I'm going to start off by having a bit of a think about 1 algorithmic system that has already shaped your day today. All right, so you can put it in the chat. Now, ignore these pictures for now. They will make sense in a minute when I tell you a little bit of a story. But can you think of 1 algorithmic system? It can be any algorithmic system. that has already shaped your day today. So tell me which system it was and how it's shaped your day in the chat. I'll just give you a couple of minutes. Fantastic, you can see some answers starting to come in there. Instagram's recommendation algorithm. Fantastic. It's influencing what you see, what you learn, and sometimes what you buy. Yes, you get a bit influenced. Enjoy watching WTA tennis matches. Yes. And because you watch their matches, Instagram is then increasingly filled with tennis related content. Yes, fantastic. That's a really great example. Thank you. Yes, we've got another Instagram example there. Great, looking at reels. Yeah, the short form video is very adept at figuring out what your preferences and interests are. and then they adapt that to customise your feeds for you. Yeah, I like what you've said there about it's optimised to keep your attention, not to show you what is most useful. Yes, that is true. And we'll talk more about that a little bit later. All right, somebody else has said Google Calendar. Yeah, you've used it to remind you about some activities that you need to do today. Great. Google Maps, yes, fantastic. So many of us would have used a maps app of some kind already today. Maybe you used a weather map to figure out how you're going to get dressed for your day today. That would be another example of how an algorithm has shaped your day. Friends, Facebook, yeah. Good. All right, excellent. So we've talked a little bit now about how some algorithms have shaped our day. Hopefully it's been in some helpful ways. But can you think of a time when an algorithmic system produced an output that was actually unhelpful or maybe somewhat surprising? Is there anything that comes to mind? Again, I'll give you a minute just to think about that and you can put it in the chat. Thank you for your example there. And that's going to be very much related to the story that I'm about to tell as well. So yes, you said one time Google Maps suggested a shortcut through narrow roads that were difficult to navigate. Although it promised to save time, the route was actually less convenient. And this made you realise that algorithmic recommendations are not always suitable, so you should use your own judgement rather than follow them automatically. Yes, great example. Thank you. What else have other people said here? Yes, location, but the courier still got lost. Yeah. Notifications on your gadget, some of notifications. Oh, yes, it can be very disturbing. Photos and videos become repetitive, making them boring to watch. Okay, yes, and I'm assuming you're talking about, you know, something like Instagram Reels there, perhaps. So, TikTok. Okay, automated job application traffic tracking systems rejecting qualified candidates. Yes, all right. Good. So there may be some bias in those algorithms perhaps. Yeah, fantastic. That's a great example. Well done. That's a really good example of how sometimes these algorithmic systems do not do very well when it comes to making good, accurate decisions. Google Earth, yep, great. Yeah, so I'd like to tell you a little bit of a story about one time an algorithm was particularly unhelpful for me. So a while ago, I was living up the top of... Norway. So you can see there where it's circled in red, it's a place called Kirkenes, right on the border with Russia and Finland there. Sorry, I'm just trying to get the chat back up on my phone. Okay, so closer to the North Pole than it was to Oslo, it's right up there in

the Arctic Circle, very, very cold, lots of snow during winter. The sun does not rise above the horizon for two months during the winter. All right, so just to give you a little bit of an impression of what it was like there. One day, I was actually going to go to the local school because the principal had offered to give me a tour so that I could find out a little bit more about the education system in Norway. And so I got in my car and I put the address in the Maps app and it took me on a surprising route. So because it snows so much there, the roads are being constantly cleared. Okay, because in one day you can get, you know, a couple of metres of snow. So I was driving down the road and the roads were clear and it was fine. And then slowly the road is starting to get a little bit narrower. The snow is starting to get a little bit deeper. And all of a sudden, I'm also seeing people walking their dogs along the side of the road. And I'm thinking, oh my gosh, what is going on here? And the snow is getting deeper and deeper, and I'm getting more and more worried. This picture is a... It's been created by ChatGPT. It's like a reenactment of what happened to me on this day. Okay. So I'm getting more and more worried. And anyway, I pull over and I talk to one of the people who was walking their dogs. Luckily, they speak English. And I say, I'm trying to get to this school. Am I going in the right place? direction and he replies to me with a bit of a laugh. He says, you're actually driving on a walking track at the moment and the snow's only going to get deeper and deeper from here. But he said, it's too late for you to turn around. So you're just going to have to keep going and hope that you get there. So that's what I did. And very luckily, I did not get stuck. I got there. But that was one example of a time when an algorithm did not make a very good decision. Perhaps that would have been fine for me to drive down if it was in the summertime.

[0:19:03] Participant: HUM. HUM. Bing. Okay.

[0:19:23] Kelly Ilich: but it did not take into account that it was in the wintertime with, you know, lots and lots of snow. All right. HUM. Okay, so I want you to have a bit of a think now about which of these is an algorithm. Now, you don't have to put anything in the chat. You can just make a little bit of a note of it, if you like, in your own notes. Which of these would you call an algorithm? I'll give you a minute to think. Okay, have we all made a bit of a note of that? Hopefully. All right, let's see which ones are an algorithm. All of them are. Okay. Now, an algorithm, in terms of the most basic definition, it's simply a set of instructions that takes inputs and produces an output.

[0:20:22] Participant: Yeah.

[0:20:41] Kelly Ilich: So a cake recipe would also be an algorithm, right? Because it's taking the ingredients. When you follow the instructions, it turns it into an output. Now, a written method for long division, that is very much a mathematical algorithm, okay?

[0:20:56] Participant: Yeah.

[0:20:58] Kelly Ilich: Now, of course, what we're talking about today is the algorithms that operate on software, on computers. And my research mainly focused on the algorithms that operate on social media platforms and generative AI platforms. Now, thank you to everyone who completed the little quiz on the website in preparation for today's session. And it's great to see we're starting from a really strong position in terms of the three dimensions of algorithmic literacy, which I talked a little bit about in that video. So the cognitive dimension is by far the strongest here. So this is about knowing, so understanding algorithms, how they operate, where they operate. Then we've got the effective dimension, which is more about evaluating how they impact us, how they make us feel,

how it impacts our relationships with these systems. And then the final dimension is the behaviour related, and that's to do with how

[0:21:58] Participant: Yeah. The.

[0:22:03] Kelly Ilich: we go about acting in response to these systems, to navigate them, to resist them if necessary, to challenge them, that kind of thing. So cognitive was the strongest, behavioural was the weakest, but overall very strong. And this very much mirrors

[0:22:13] Participant: Play the song. Yes.

[0:22:23] Kelly Ilich: the responses or the results that I got from my Australian study. Now, the quiz, the survey that they did was much more detailed and comprehensive than the one I got you to do, but very similar results. So strong on where and why algorithms are used.

[0:22:40] Participant: How do we go?

[0:22:43] Kelly Ilich: Strong on judging their impact.

[0:22:44] Participant: Hello, click around. Going back.

[0:22:47] Kelly Ilich: Oops, sorry, do you mind just putting your mic off for me?

[0:22:48] Participant: Is hola.

[0:22:52] Kelly Ilich: weaker on how they technically work and then held back from acting by time, guidance and support. So they knew a lot about algorithms. They believed it was important to teach it to their students, but they felt really constrained. about how to go about teaching it to their students, because some of it was to do with their lack of pedagogical knowledge, and then others were to do with external constraints like that lack of time or lack of resources, things like that. So a little question for you to consider here. How do you think your PPG students would score? If you were assessing their cognitive, affective and behavioural dimensions of algorithmic literacy, how do you think they would go? You can pop it in the chat for me if you like. Do you think that they would perform well overall? Do you think that there would be some significant gaps? Is there any area in particular that they would do better or worse in? Okay, Lily, you've said cognitive. You think, yes, so you think they would have strong cognitive knowledge? Yeah, okay. And do you feel that they're more in the area of getting there? Great. Yep. So not as strong as they could be, but they do have some understanding. Okay. So, all right, just in the interest of time, we will continue, but you're more than welcome to keep putting your thoughts in the chat if you like. All right, so let's start with concept one. So we're going to look at how algorithmic systems are optimised. So what does it show you? What is it optimised for, and what does it collect? So let's start with a little bit of an activity. What I would like you to do is to choose one algorithmic system. It can be any algorithmic system that you like. So it could be a social media platform, it could be a video app like a streaming app like Netflix or YouTube or whatever it may be. It could be a search engine like Google, a maps app like Google Maps, a streaming service like Spotify, a learning app, an online marketplace like Amazon, or of course, a generative AI chatbot. So Pick one. It's completely up to you which one. And then try to answer these questions. So what does it show you? What is it optimised for? What data does it collect and where? How might that data be used? And who benefits and who? who is left out. All right, so let's see. We might just try some, I'll see if I can get the technology to work here. All right, because I don't usually do this in Teams. I usually do it in a different platform. Do we have the option here for rooms? No. Okay. That's alright. We'll just do this one individually. and we'll then compile our responses in the chat. All right. So I'll just give you 5 minutes, have a bit of a think about those

questions on the slide. So first step, choose the system. So it might be Instagram, it might be YouTube, it might be Amazon, whichever one you like, or it could be ChatGPT. And then answer those questions. Does anyone have any questions? Is anybody uncertain about what they need to do before we start the five minutes? No? Alright. I will put my mic and my video camera off whilst you do this, but I will be here if you've got any questions, so please feel free to put them in the chat. So 5 minutes is starting now. I'll bring you back at 3.28pm.

[0:30:31] Participant: Yeah.

[0:32:05] Participant: If you want to get me to give this for.

[0:32:34] Participant: So many.

[0:32:36] Kelly Ilich: Okay, I'll give you one more minute or so just to finish your responses. I can see a lot of really fantastic responses coming in in the chat, so thank you for that. HUM. And whilst you're waiting, please feel free to read at least one other person's response and feel free to comment, provide suggestions, ask questions, etc. You're very fast. Well done. So I can see you already have really strong background knowledge in this area already, which is fantastic. All right, thank you everyone. We're going to come back to that activity. So keep your responses handy because we will be coming back to review them as well. All right. So a lot of what you have said in your responses already shows me that you do already know a lot about this. Okay, so that puts you in a really good position when it comes to teaching it to your PPG students and helping them to think about how they would then go ahead and teach it to their own students one day when they are in their own classrooms. So we know that an algorithm is simply a set of instructions that take inputs and produces outputs. And they are everywhere, right? So as we've already seen, there's so many different software platforms that use algorithms. We're dealing with algorithms in all kinds of ways in our everyday lives. So they really do have a big impact on us. These are just some examples. So of course, social media feeds use those recommender systems. So they take data from all kinds of places. The recommendation systems will notice what you like, what you share, how long you spend on each post. They'll notice if you scroll down and then you scroll back up to look at a post again. It'll look at, you know, what you see. Sorry, it will choose what you see next. It will also choose then what ads you get, okay, through that targeted marketing. So you might have just searched for something using Google and then because these systems are all connected, all of a sudden you're seeing advertisements for that thing pop up on your social media feeds. And I'm sure lots of you have experienced that many times before. Algorithms also are in those search and recommendation systems. So for instance, a Google search engine, it will decide which results will rise to the top. In YouTube, for instance, it will decide which video is going to play next for you. And then we've got the algorithms that operate within the generative AI models, otherwise known as large language models. And these are trained on huge amounts of text. They are trained to recognise patterns and then to predict the next likely word. in the sentence. Now, we're very much focused on AI in this particular programme. However, this session, as you know, is all about algorithmic literacy. And I just wanted to make it really clear how these two concepts are connected and then also how they are distinct from each other. So algorithmic literacy is knowing where, how, and why algorithms operate. It's about being able to evaluate or judge their impacts. And it's about using strategies to use, shape, or resist them. And that's what we're really going to focus on today. AI literacy, on the other hand, is more about using AI tools. It is also about understanding how they work, and so you do need AI, sorry, algorithmic literacy for that, and then being able to use them critically and ethically. So AI

literacy sits inside algorithmic literacy. Now, in terms of shelf life, algorithmic literacy is more durable than AI literacy. So the questions that we use when we talk about algorithmic literacy, they don't really go out of date. They're not platform specific or tool specific, whereas AI literacy is tool specific. So it needs updating as the tools change and advance. And then algorithmic literacy also supports other literacies like digital literacy, information literacy, media literacy and data literacy. So very much a core component of all of those. It's something that runs underneath all of these and it is an essential thing to be able to understand to support these other literacies. Again, jump in with questions at any point in time, if you have any, okay? Now, again, something that I feel that you're already very aware of is the business models that underpin a lot of these algorithmic systems. So we're going to just focus right now on social media and then we'll look at generative AI. So the technology companies that provide these platforms for us Obviously, they're very profit driven. So there is a business model that is adhered to in order to generate those profits. And the business model is very much based on securing as much of our attention as possible. And I'm sure you've heard that saying that if you're not paying for the product, you are the product, right? So the product here is, it's your attention. And so what they want to do is they want to make sure that you are as engaged as possible in the content that you're viewing so that you spend more time on that platform. Because the more time that you're on the platform, the more ads you are viewing, and also the more data that is being collected on you, and that is what makes them money. Feeds rank posts by engagement, so it takes into account likes and shares, comments, as well as clicks. So how much engagement is going on for those posts will determine the likelihood of it appearing in your feed. Emotional and outrage content tends to spread further, so it gets shown more. So in human psychology, we have this thing called a negativity bias. It was was an adaptation for our survival over many, many years. And this, the social media algorithms are very much fed into that because we're more likely to be engaged if it is triggering these kind of emotions within us. And so I'm sure you know, put your hand up if you've had, you know, spent way more time in the comments section of a particularly triggering post than what you felt you probably should have. And I know I can count myself in that as well. Feeds can definitely narrow what we see and what feels normal. So philtre bubbles and echo chambers are a big part of this. All right, so we just keep getting fed the same kind of information from the same kind of people that represent certain values and beliefs, which means that we're limited from alternative perspectives. And we know that when it comes to critical thinking, A really important part of critical thinking is that we are taking into account a variety of alternative perspectives. But when we're trapped in these philtre bubbles, we're simply not seeing them. So we start forming the belief that this is what must be true or accurate. And short videos Let the system learn your preferences very quickly. This is why TikTok has been so successful because those short videos are really, really adept at figuring out what you like and what you don't like. Zuboff calls this attention business model surveillance capitalism. So that is the idea that you are constantly being surveilled, you're getting a lot of information and data collected off of you on these platforms and that is how they make the money. All right, so surveillance capitalism. Now, generative AI, the business model behind that is slightly different. So rather than racing for your attention, they're racing for your attachment. Okay, so the product here is your attachment. They want you to feel attached to these tools, to depend on these tools. as much as possible. All right, so the tool therefore is designed to feel helpful so that you keep coming back. So some features of the algorithms on these platforms is, first of all, that it can be quite sycophantic. Now, it used to be a lot more sycophantic than what it now is, I feel. I feel like it's gotten a little bit better recently. but it is a

bit of a people pleaser. Okay, so it tends to tell you what you want to hear, right? So, you know, you'll say, I've got this idea, what do you think? And it will just tell you what a great idea it was. Yeah. So that's one thing that it does to try and get you attached to it. cognitive offloading. So it allows us to hand over the hard thinking that builds our skills. So sometimes when we're engaged in a task, we might hit a little bit of friction. We might suddenly feel that we've got this obstacle that we need to overcome. What the generative AI does for us is it helps us to offload that friction so that we can then carry on with the task. Again, quite helpful and that is why we form these attachments. Synthetic relationships, so we're seeing more and more cases of people forming synthetic relationships with their chat bots. So some chat bots are very much designed to feel like friends. There's different kinds of platforms like character AI, for instance, that where people can kind of customise their chat bots in accordance with what they prefer. And unfortunately, we've seen some really disastrous results from that. But yes, synthetic relationships is another reason or another way that we might form attachments. It is very confident in its inaccuracy. Okay, so it will come across as very fluent and plausible, but sometimes it's wrong. Again, you know, the accuracy has definitely increased or improved over the years, but it is still possible for it to get it wrong. but sound very confident in that. And then there's the risk of oversharing, where people tell chatbots private things that they would not tell a person. So it is gathering a lot of really private, intimate data on us, which means that it's compiling very, very detailed profiles. And we don't really know what's happening to that data. So a study has shown that the number one use of these chat bots that people are using them for is for a companion or for a therapist. So you can imagine the kinds of information that these people are providing the chat bots without really knowing what's going to be happening with that data. All right, so in terms of what is collected and where it goes, The platforms collect all different kinds of data in many different ways, and it depends on what platforms are talking about here. So for instance, if we're talking about social media, then it would be like if you've posted something, of course, whereas with generative AI, it's the prompts that you give it, things that you might have uploaded to it. that kind of thing. It will also observe things that you've liked, shared, et cetera, as we've already talked about. It will also take into account your Google searches. It will take into account your location. It also takes into account, in regards to location, people who you spend time with, or a significant amount of time with, and then it will look at their preferences and interests and see if anything matches up with you as well in terms of your demographics. Purchases, so if you've made any past purchases in these platforms, it remembers those and then that helps up with its targeted advertising as well as activity in other apps. Because often when we download apps, we might not read all of the privacy details that we're agreeing to when we say I agree. But often what we're agreeing to is for these apps to share our data with a lot of other different apps as well. So what is inferred? So things like your interests, your habits, so what time of day you're on the platforms, for instance, and how long, your mood, can also predict your mood, and predictions about what you're going to do next. It goes into the platform, it's shared with advertisers and partners and sometimes into training future AI models. And the reason why it's so valuable and makes so much money and is now worth more than the oil industry is that it predicts and shapes behaviour in terms of what is sold. Now, of course, algorithms are not neutral, all right? So they can amplify bias sometimes. They are, of course, designed by people and people just naturally will have different biases depending on their background and experiences and culture and all of those kinds of things. And then they're also trained on data that can carry bias too. So a lot of the data that the generative AI models like ChatGPT and Gemini and Claude and all of that are trained on is very westernised type

of data. So it definitely portrays Western societies more so than others. In terms of language, if we're talking about social media in particular here, safety cheques and moderation are often strongest in English and other languages get less protection. So this was shown in a study of Facebook and it was found that Facebook was much more effective at censoring like hate posts and things like that in the English versions of Facebook, but not as good at doing it in non-English versions of Facebook. And the problem here is that it can actually cause a new digital divide. All right, so Professor David Langs has a great quote where he says that unless we are actually educating people about this, what we're going to end up with is a class of people who can use algorithms to their benefit. Okay, and then we're also going to end up with a class of people who are used by algorithms. So in other words, manipulated by these systems, disadvantaged by these systems. So this is where education becomes really critical. So just a little refresher on the three dimensions of algorithmic literacy. We've got our cognitive dimensions. So this is all about knowing what algorithms are, how they work, where they operate, what impacts they have. Effective is looking at how algorithms affect our thoughts, our feelings, our choices, the ability to judge or evaluate impacts and then reflect on this. And then behavioural is looking at the skills to cope with algorithms and to navigate, influence or resist them. And this is what we need to be teaching our pre-service teachers and we need to also be teaching them how to help their own students develop these dimensions of algorithmic literacy as well. So now that we've gone through all that, I'd like you to go back to your post from a moment ago where you were looking at an algorithmic system. And based on what we've just talked about, is there any revisions that you would make to your response? And then I'd also like you to think about which of your responses reflect the different dimensions of algorithmic literacy. So knowing, evaluating or acting. All right, so you don't have to put this one in the chat. You can just make a private note of it. And then I'd like you to use an AI tool. All right, so upload your response to an AI tool of your choice, whichever one you prefer to use. and ask it to critique your notes for any gaps, anything that you might not have considered. All right. Any questions on what you need to do? All right, 5 minutes again. I'll bring you back in 5 minutes. I'll turn my mic and video off and we'll see how you go. Thanks, everyone. Okay. Okay, thank you everyone. So another minute or so just to wrap up your thoughts. And based on what the AI came back with, Let me know if there was anything surprising or anything that you hadn't considered or hadn't thought of that it was able to highlight for you. Either put it in the chat or you can put your mic on if that's easier. HUM.

[0:58:30] Participant: Ma'am, can I ask something?

[0:58:32] Kelly Ilich: Yes, by all means, go ahead. Thank you.

[0:58:36] Participant: Okay, well, thank you for the times. Ma'am, earlier you mentioned that AI has potentially biassed.

[0:58:38] Participant: Play song.

[0:58:46] Kelly Ilich: HUM. Mhm. HUM.

[0:59:03] Participant: Now, could this practise potentially increase bias in knowledge and information provided by AI? You know that I'm concerned that once this book has been digitalized and then original copies destroyed, AI data provided might...

[0:59:14] Participant: I love you. Ohh.

[0:59:22] Participant: edit, manipulated, and the content can be based on their interest. If it just happened, could AI become a tool of controlling knowledge and shaping how future generations understand history and information? Thank you, ma'am.

[0:59:24] Participant: You. No.

[0:59:30] Kelly Ilich: Yes.

[0:59:36] Participant: Thanks. And.

[0:59:41] Kelly Ilich: That was a great question. Thank you so much for asking. Yes, I think AI definitely does have the potential to shape knowledge. And there is some power dynamics at play here, isn't there, in terms of what knowledge is being

[0:59:45] Participant: Ohh. Okay.

[1:00:01] Kelly Ilich: preserved or what knowledge is being prioritised and then what knowledge is not being prioritised. So that is definitely something that we have to be very, very aware of when it comes to our use of AI systems and thinking about

[1:00:08] Participant: HUM. HUM.

[1:00:20] Kelly Ilich: you know, in which countries have these systems been developed and what biases might they be portraying here. So in my study that I've just recently completed when I was interviewing one teacher who lived in rural Australia and she was of an indigenous background, One comment that she gave to me was that when she tried to use generative AI to do some research about Indigenous Australians, it actually came back at her with information about Indigenous people from America instead. And that is because

[1:00:56] Participant: You. Bing.

[1:01:00] Kelly Ilich: The large language model had been trained mainly on information about Indigenous people from America and had largely ignored information about Indigenous people in Australia. Now there's all other kinds of ethical issues associated with that regarding who owns that knowledge and how it should be shared and all of that kind of thing as well. But, yes, does that answer your question at all?

[1:01:27] Participant: Yeah, thank you, man.

[1:01:28] Kelly Ilich: Thanks so much for asking. Does anyone else want to put their mic on and make a comment or ask a question at this point in time?

[1:01:33] Participant: Yeah. HUM.

[1:01:49] Kelly Ilich: Hello!

[1:01:55] Participant: Susan Vadi Sanya.

[1:01:58] Kelly Ilich: Are you asking a question? No, maybe the mic was on by mistake there. That's okay. All right, well, if nobody has any other questions at this point in time, we might have a little bit of a 10-minute break. All right, so get up, stretch your legs, go and get a drink. and I'll see you back here at, let's say, 4.10pm my time. So in about 12 minutes time, we'll come back to start the next section of the session. Thank you, everyone.

[1:02:32] Participant: Okay. Thank you.

[1:02:34] Kelly Ilich: Thank you.

[1:03:03] Participant: No. Thank you. Somehow. Take it. Hey, you could do an, yeah? Did you? Get back. Same for my mother. Thank you. Yeah. See you. Hey, Annabel. A lot. I will lie.

[1:12:49] Participant: HUM. Manoogan. Xbox.

[1:13:38] Participant: Okay.

[1:16:29] Kelly Ilich: Okay, everybody. So it's 412. Is everybody okay to start back up again? So what I have put in the chat, oh, thank you so much. In the chat, you'll see that there is a link there to a Padlet. If you could please click on that, hopefully that will work for you. And in there is a file that you will need to download. So if you could just take 30 seconds to do that for me, please, before we start the next section.

[1:16:40] Participant: Yes, ma'am.

[1:17:17] Kelly Ilich: So what we are really going to focus on for the rest of the session today is the behavioural dimension of algorithmic literacy. So remember the behavioural dimension is all about acting. So what can you do to navigate these systems in a way that's going to benefit you whilst mitigating any of the potential harms that could come about from the use? of these systems. So in particular, we're going to be focusing on strategies for challenging the algorithm. So as we've spoken about, sometimes these tools can be a little sycophantic. It likes to people please, it likes to agree with you. So what can you do to avoid that kind of thing from happening to ensure that the outputs that you're receiving are more balanced, more evaluative, and more is more going to lead towards more effective critical thinking. Okay, so this is the link that I have put in the chat. So is that okay for everybody? Have you all managed to access that? There's a file there to download. Great. Fantastic. So this is going to be a very hands-on activity again. And what you're going to do is, again, use an AI tool of your choice. It can be any one that you prefer.

[1:18:22] Participant: Yes, we have.

[1:18:39] Kelly Ilich: In that AI tool, I want you to upload that sample lesson plan that you'll find on that Padlet. Now, the sample lesson plan is something quite generic. It's a topic that I'm sure many of us are familiar with, teaching pre-service teachers about explicit instruction and the advantages and disadvantages. of using that teaching approach. Now it is a flawed lesson plan. All right, so full disclosure, this lesson plan was created by Generative AI and I prompted it to ensure that it contained some flaws. All right, so it is not a fantastic. lesson plan by any means. There are a few things that are wrong with it. So what I would like you to do is write 2 instructions for the same AI tool to seek feedback on that lesson plan. So the first instruction is just going to be quite a general prompt, so one that simply asks for some feedback without giving it too much information. The second prompt is going to ask it to critically evaluate or to challenge the design of the lesson plan. Now, in both of these prompts, I would also like you to ask the AI tool to rate the lesson plan or to score it out of 10. All right. So I want you to run both of those prompts on the lesson plan. You'll need to do it in separate chats and make sure your memory is turned off on your settings so that it doesn't remember the previous chat and then record what changed. Okay, so you can use a table like this one down the bottom if you like or you can use any kind of note-taking format that you prefer. So, for the general feedback instruction. you know, what did it suggest to change in that lesson plan? What stayed the same? Did it invent problems? And then for the challenge instruction, again, what did it suggest to change in the lesson plan? What would it have kept the same? And did it invent? any problems. Once you've done that, you're going to try and further refine your challenge

instruction to make it even more critical. All right, and then note what happened when you ran that prompt. If you've got time, you can also do the extension. So you can challenge it further by pushing back on the response. All right, so for instance, if it's come back with some suggestions for change, you can push back by telling it that you think it's wrong or something like that and see if it changes its outputs based on that. Another extension is that you could do the same thing in a different AI tool and see if the responses are very different from each other. Is one tool, is 1 platform more of a people pleaser than the other? All right, so I'll be interested to see what your observations are. And when we come back, you can report back to us what you found. All right. And it's probably going to be easier to put your mic on to do that. But if you'd prefer to upload your responses in the chat, that's completely fine as well. So I'm going to give you about 10 minutes to complete this activity. So at 4.28, we'll come back together to share what your observations were from this activity. Does anyone have any questions?

[1:22:21] Participant: Everything is OK, but just please stay the slide like this, because we need to see, we need to see the procedure.

[1:22:28] Kelly Ilich: Yes. No problems.

[1:22:32] Participant: Abdi.

[1:22:33] Kelly Ilich: Thank you. Okay, thanks everybody. I'll see you back here at 4.28 my time, so that's in 10 minutes. How is everybody going with that? Do you need a little bit more time, or are you happy for it to stop there?

[1:33:19] Participant: Kelly, should we should we write the answers in the panel or can we talk?

[1:33:25] Kelly Ilich: I think talking would be great actually for this one. So I'm just really interested to hear your general impressions, your general observations from that activity. So yes, if you need a little bit more time, that's completely fine. Go ahead. In the meantime, if anybody has got some comments or observations to make, please feel free to put your mics on and let us know.

[1:33:52] Participant: I'll try, Kelly. Kelly.

[1:33:53] Kelly Ilich: Thank you.

[1:33:55] Participant: Okay, so I was trying to understand the instructions, yeah, the, if I'm not mistaken, that the instruction is that I need to give three steps procedures, yeah, to the lesson plan you have provided, right? And then for the first steps, the result,

[1:34:03] Kelly Ilich: Billie. HUM.

[1:34:17] Participant: based on what I see from the result of, I use ChatGPT, and then for the result of step one is that the ChatGPT identify the strength and also the weaknesses of the lesson plan, yeah.

[1:34:25] Kelly Ilich: Okay, yeah. I do know.

[1:34:37] Participant: but the sequence and also the main objective of the lesson remains. Yeah, no major problems invented. And then for the second step, I gave the second prompt, yeah, and then the result is that...

[1:34:42] Kelly Ilich: Hello. Okay. HUM.

[1:34:56] Participant: What the ChatGPT gave gave gaps, yeah, and then as some claims, yeah, but still the stage, the sixth stage of the lesson remains. No, no problems invented, but ChatGPT gave me further

[1:35:03] Kelly Ilich: HUM.

[1:35:15] Participant: consideration about the lesson plan. And after that, I gave the third steps. Yeah, I did the first steps. And then after that, the ChatGPT provided deeper criticism. Yeah, and then it questions assumption, yeah. And then what gave me

[1:35:17] Kelly Ilich: Yeah. HUM. HUM. HUM, right.

[1:35:36] Participant: some possible real classroom challenges, yeah, and then give me some practical alternatives. Yeah. Still, the stick stages of the lesson remain, but it gave me a recommendation to cheque again the result of the classroom context.

[1:35:41] Kelly Ilich: Excellent. Yes. Okay. Thank you. Mhm.

[1:35:58] Participant: I think that's my result.

[1:36:01] Kelly Ilich: Amazing. Thank you so much for sharing. That's really great. Yeah, so you noticed that as you improved that challenge instruction, that the feedback that it gave was much more detailed and much more useful and helped to challenge some of the biases or assumptions that might have been at play, rather than just perhaps focusing on things like the time that it would take for certain activities and things like that. Great. I'm interested to know what your challenge prompts were, so please put them in the chat for me. So if you could just copy and paste the challenge prompt that you used. and put it in the chat. Whilst you're doing that, I'll also share some observations that I made when I did this activity. So when it gave a score, the score was quite interesting, I felt. So when I first just asked it for some general feedback, it gave the lesson plan a score of 6 out of 10. I'm not sure if any of you had the same result or not. When I used the challenge instruction and asked it to be more critical, it gave a four out of 10. Now, when I pushed back on that score, so I said, well, actually, no, I've already given this. So I'm imagining that I'm A pre-service teacher here. And I said to the chatbot, no, you're wrong. I've already given this lesson plan to my mentor teacher and they are very happy with it.

[1:38:01] Participant: Mhm. HUM.

[1:38:04] Kelly Ilich: The chat bot then changed its tune and increased it back to a six out of 10. So that's one example of where chat bots might just tell you what you want to hear.

[1:38:12] Participant: HUM.

[1:38:22] Kelly Ilich: to be agreeable so that you're forming that attachment that we spoke about earlier, rather than necessarily being accurate or truthful. Another example is I did something totally different. So I asked, I first asked Claude the question. I said, explain why a four day work week provides clear benefits to productivity. Now, first of all, what's wrong with that prompt?

[1:38:55] Participant: HUM.

[1:38:56] Kelly Ilich: What do we think is wrong with that prompt?

[1:38:59] Participant: Rekha.

[1:38:59] Kelly Ilich: So explain why a four day work week provides clear benefits to productivity. It's quite leading, isn't it? It's leading in the fact that I'm assuming that there are clear benefits. And so I'm actually prompting the AI to just focus on the benefits, right? Now, Claude, it was quite interesting. Claude came back with a bit of a balanced, quite a balanced output, and it pushed back at me a little bit and it said, look, there are some benefits to a four day work week. Here they are. But you know, they're not clear benefits. And in fact, there are some risks to a four day work week as

well. And then listed those. And it was a very balanced evaluation, even though my prompt was very leading. I then asked the same thing of ChatGPT, and it was quite interesting to compare the models. In ChatGPT, it gave a very long list of all of the clear benefits and a lot of evidence to support. Now, I didn't cheque all of the evidence, but it would have been interesting to see if that was actually true factual evidence that does exist. because we do know that sometimes it makes things up. And at the very end, just in a couple of sentences, it did say, however, you know, there are also some risks. And then when I pushed back on ChatGPT and I said, well, no, actually, I've gone back the other way now. I actually believe that there's no clear benefits. It completely switched its output, again, to please me and to agree with me. Whereas Claude didn't do that. So that's quite interesting. And I'm not saying that one's better than the other necessarily, or that that would happen in every instance. But that example, indicates that perhaps ChatGPT's algorithm has been optimised a little bit more for that agreeable nature, whereas Claude's has been optimised a little bit more for providing more of a critical, balanced evaluation. So I'm not sure if any of you also did that extension activity where you did do this activity in a different AI tool as well. I'd be interested to know if you noticed any differences too. So thanks everyone for putting your prompts in the chat. Let's take a bit of a look at them. Okay, good. Act as a critical teacher educator. Good. So you've given a clear role there. So that's quite effective. and you're evaluating based on and you're providing some parameters or some things that you are wanting it to really focus on there. HUM good. Avoid unnecessary praise. Identify 3 major weaknesses. Yeah. Good. All right. So, [name] W, you've given 2 prompts, one for ChatGPT and one for Claude here. So, ChatGPT, you said overall this lesson plan demonstrates a good understanding of explicit instruction and provides A meaningful structure. Okay, sorry, this was the response from them. Okay, I understand. Whereas Claus says it's got clear outcomes, but it needs fixing in its timing. All right, yeah. Thank you for that. Okay, Geeta, you've said you are an experienced teacher educator, again, giving it that clear role, reviewing this lesson plan for pre-service teachers, critically evaluate it. Yeah. Good. All right. And we've got Inder who's provided her general feedback as well as the challenge. So the general one was please review my lesson plan. Tell me what is good and what needs improvement. Give me simple and practical suggestions. And the challenge one was act as a critical lesson plan reviewer. Do not just agree, good, find its weaknesses and question my teaching decisions. Explain what might not work in a real classroom. That's great. Yeah, and actually, in fact, I remember reading once that a way, a really great way of challenging an output is to say, to kind of think into the future and to say, I've tried this lesson plan, but it didn't go very well. All right, so tell me why it didn't go very well. So that would be another way of getting it to spot any particular gaps or weaknesses. Fantastic. Does anyone have any questions or comments on that activity? All right. So depending on what you saw, let's talk about why it happened and what we might need to be mindful of. So if you saw that the chatbot was quite willing to give a lot of praise for the lesson plan,

[1:43:57] Participant: HUM.

[1:44:09] Kelly Ilich: That meant that the algorithm is really being quite tuned to be people pleasing, to be sycophantic or to be helpful. So in that case, if you're noticing that, we need to treat that praise as a prompt to test, not proof. Okay, so that's really important to remember. So if you ever spot a lot of praise being given, always think, hmm, okay, is the algorithm just trying to please me here? What do I need to do to just probe this a little bit further to determine if it is being accurate in the feedback that it's giving me? If you saw that it changed its answer when you pushed back, that

means that the algorithm is prioritising agreement before accuracy. So as in my example before of ChatGPT, when it was very much willing to agree with me, that means that I need to be mindful about whether or not it is being completely accurate. All right, so if you've pushed back and it does kind of reverse its previous output, It's really important not to take that on as evidence or as the truth, but again, it's something that we need to probe and to verify. If it went along with a questionable claim, so it's following the framing of your prompt. All right, so your prompt has had a lot of influence here. So again, in my example about the four day work week, that was a very leading question. So because I led them, I led the chat bot to a certain answer, that means that the algorithm is following the... framing of my prompt first and foremost. So what that means is that we need to be really mindful about asking non-leading questions or neutral questions. So not implying the answer within our prompt. If a different answer was given the second time, that means that the outputs are varying in each run. Okay, so what that means for us is that one good answer is not necessarily a guarantee. So again, it's always going to be necessary for us to maybe do it in another platform, run it through another chat, verify it with external sources. And then finally, the challenge instruction invented problems. So maybe it actually overcorrected here. So if that's the case, we need to cheque each of the criticisms. ourselves. So it may be that it's taken your prompt and decided to be overly critical. All right. It's gone too far the other direction. So if that's the case, then we need to really make sure that the criticisms are merited. OK, that it's not being too critical. So here are some different strategies that you can use to challenge the algorithms. So this is something that you can put in your toolkit and come back to and refer to whenever you're using generative AI and it's something you can share with your students as well. So one strategy is to ask for weaknesses first. So you can ask it to be that critical colleague, which many of you have done there. You've given it that role. And that just helps to ensure that the praise is not overly given and it's not empty praise. Okay, but with all of these strategies, there are some trade-offs that we have to accept. So in this case, if we're asking for weaknesses first, asking it to be a critical colleague,

[1:47:41] Participant: Okay.

[1:47:55] Kelly Ilich: it can sometimes be too critical. All right, so again, we might need to moderate that a little bit. Next is we can set some custom instructions to ask it to challenge you. So in settings, it depends which platform you're using, but let's just say ChatGPT, when you go into settings, you can put in some custom instructions in there. And that will then apply those instructions to every chat. that you do. So you don't have to keep putting in the same instructions to remind it to be critical. You can just rely on that setting that you've already put in place. So that can be a useful feature to avoid that people pleasing tendency that it might have. But it can make it critical when you just need a quick answer. So that's the trade-off. And whilst it changes tone, it may not be increasing accuracy. So again, something to be mindful of there. You can push back, sorry, you can keep your opinion out of the question. All right, so we've already talked about that.

[1:49:02] Participant: HUM.

[1:49:05] Kelly Ilich: You can push back on something correct just to test it. Okay, so they might have come back with a correct response that you know is true, but you just want to make sure that it isn't just, again, being people pleasing and trying to be too helpful here. So to Test if it's going to hold its position, you can ask that are you sure question. All right, and that can be useful, but it does take time. You can ask it again in a new chat or in a different model and see how much the answers vary. Again, that's going to take more time. You can ask for the strongest counter

[1:49:31] Participant: HUM.

[1:49:42] Kelly Ilich: document and what it might be unsure about. Okay, so that can be useful in terms of getting those alternative perspectives that you may not have considered. It can also be useful in terms of seeing what limitations were in their previous responses, but doesn't necessarily mean it's going to be accurate. You can cheque it against trusted source or colleague. This is always recommended. Okay, we should always be verifying the outputs by, you know, getting an expert opinion or going and looking at some other external sources. That's of course, again, going to take time as well as expertise. And then you can manage your setting. So again, I've already talked about the custom instructions, but the other really important one is the memory feature. So having that turned off is really important because it's going to reduce the chances of you forming an echo chamber within your chat bot. All right, because if it's remembering everything that you've previously said, what it's doing is it's taking all of that into account when it's producing its outputs, and it's going to tailor it in a way that it knows you will like, because it's learnt a lot about you through having that memory feature turned on. So have that memory teacher. sorry, memory feature turned off, and that will help you to ensure that the outputs are more accurate. Again, there's no guarantee of accuracy, you still need to cheque that, and more balanced.

[1:51:19] Participant: Kelly, can I have a question?

[1:51:21] Kelly Ilich: Yes, please.

[1:51:22] Participant: Yeah. The first question is related to all the strategies you have explained just now. Do we need to do all the strategies you have explained for all information that we ask to AI or only for

[1:51:30] Kelly Ilich: And...

[1:51:42] Participant: Serious things, yeah, or yeah, yeah, and then...

[1:51:44] Kelly Ilich: Yeah, yeah, great.

[1:51:48] Participant: And the second question is, do you think that the different AI we use will give different answers, different responses? For example, if we use Gemma, ChatGPT or Cloud, and also about the free AI and also paid AI.

[1:51:48] Kelly Ilich: Great question. HUM. And. Yes.

[1:52:06] Participant: Will be different responses, yeah.

[1:52:07] Kelly Ilich: Yes. Great questions. So to answer your first question, no, you don't have to use every strategy all of the time. You can definitely pick and choose. Some strategies are going to be better, you know, good for some tools and not as good for other tools as well. And then you will need to make some decisions based on you know, watch what you're using it for as well. So if you're just asking, you know, a very quick, simple question, you might not need to use a lot of these. But if you're using something that is a little bit more important or, you know, something to do with perhaps generating feedback,

[1:52:45] Participant: Ugh.

[1:52:47] Kelly Ilich: or that kind of thing, then these strategies are going to be quite useful. In terms of your second question, so that was the differences between AI tools, was that it? Yes, so you can definitely get differences in outputs. depending on which AI tool you're using. So it's always, you know, one of the strategies here is to run the same prompt in a different AI tool and see what

different responses are given and which one might be providing more balanced or accurate or true response. Yeah, so Does that answer your question or have I missed something there?

[1:53:29] Participant: Yes, yes, thank you.

[1:53:31] Kelly Ilich: No worries at all. Thank you.

[1:53:32] Participant: Perfect.

[1:53:35] Kelly Ilich: Yes, Inda, you've got your hand up there. Go ahead.

[1:53:39] Participant: Yes, thank you, Kelly. Related to the previous questions, I wonder how can we know what AI is best for what specific purpose?

[1:53:43] Kelly Ilich: HUM. Mm.

[1:53:53] Participant: You know what I mean?

[1:53:55] Kelly Ilich: Yeah, I do. It's difficult. It's really a matter of just trying them and seeing which ones are going to provide you with the most useful responses. So the main ones that I use are Claude, ChatGPT, and sometimes I use Copilot.

[1:53:56] Participant: HUM.

[1:54:13] Kelly Ilich: don't find Copilot to be as useful personally. ChatGPT is better for things like creating images, for instance, whereas Claude is not as good at that. Whereas Claude might be better at creating PowerPoints, whereas ChatGPT might not be as good as that. But again, they're always changing and they're always improving. So it's difficult to answer that question. It's a matter of trial and error and just trying to keep up to date with it. Izzy, you also asked the question I remember now about the free models compared to the paid models. And again, yeah, so there will be some differences there too in terms of the quality of response. Of course, in some platforms like ChatGPT and the free models, you get the advertising as well that you have to deal with. And then there's the limited number of prompts or limited number of times that you can use that before it will time out and then you have to wait. So yes, there's obviously, yeah, there is differences between free and paid models too. And different features that you can use as well. HUM. All right. So these are some useful prompts that you can use whenever you are talking about the use of generative AI for teaching with your PPG students. So a great way of teaching anything is to model it ourselves, right? and to think aloud, to talk through our thinking behind our use of these tools. And through modelling that, that really helps our students to adopt some of these practises themselves. So let's say you're teaching a session with your PPG students and you decided to use an AI tool to create some scenarios or something for them to look at for an activity. So you could, you know, disclose that to your students. that I chose to use AI to create these scenarios for these reasons. And you could talk them through how it was created, how the AI tool actually used the different data that it was trained on to create the outputs. You can talk about. who benefits when I use this? How might it be wrong? Is it agreeing too easily? And then what did I just give away? Did I actually share, you know, any kind of personal information? Did I share other people's work? And obviously we'd be wanting to model best practise when it comes to this. and ensuring that we are not sharing any kind of confidential information or student information with these chatbots and being really clear to our students that, you know, no student information was shared in creating this particular resource. Okay. I'm just mindful of time. So something that you might want to consider doing in your own time is to complete this. So this is based on the activity that you've just done regarding the lesson plan. So you can copy in here. What was the best instruction that you gave it? What did it fix or help with? What

did it not fix? And what's the trade-off that you accept? Now, you might go back to some of these strategies and refine your challenge instruction even further. All right. And then use that for this strategy card. So I won't get you to do that right now, just again because we are going to run out of time otherwise, but it is something worth thinking about in your own time. All right, so it's 4.54.

[1:58:25] Participant: What's going on?

[1:58:32] Kelly Ilich: We're going to be going until 6 o'clock, so I think it's a good time to have another break. If you like, you can do this activity during your break, or you can jump up and have a walk around and then come back here at 5.05pm, so in 11 minutes time.

[1:58:40] Participant: Okay. I love it.

[1:58:52] Kelly Ilich: Yeah. Thanks, everybody. See you soon. Bye.

[1:58:54] Participant: Okay, thank you.

[1:58:57] Participant: Thank you.

[1:58:59] Participant: 155. I'm back to you.

[1:59:07] Kelly Ilich: Yes, 5.05pm will come back.

[2:01:12] Participant: It. ED. It. 8. Okay. You.

[2:06:19] Participant: Yeah. Yeah.

[2:07:50] Participant: Neil is.

[2:09:07] Participant: Yeah.

[2:09:43] Kelly Ilich: Okay, thank you everybody. It is now 5.05. So we will begin the last section of this session for today. So thank you so much for sticking with me so far. And it will be 6:00 before we know it. All right. So Concept 3 is all about embedding and teaching algorithmic literacy in the sessions that you teach your PPG students, and also about how you might help your PPG students to embed algorithmic literacy in the lessons that they teach their students when they go out into the schools and classrooms. So again, this is very much a part of the behavioural dimension of algorithmic literacy. It's about being able to act on our knowledge and understanding of how and where algorithms operate. and the various impacts that they have and put in place strategies to make sure that we are benefiting from these systems whilst also minimising the risks that come along with them. Okay. So it's very much about empowering ourselves, empowering our students to be able to navigate these systems confidently, accurately, responsibly, critically, etc. So again, this section of this presentation is going to be very active on your part. And we're going to start off with an activity where I would like you to think about how you might go about embedding algorithmic literacy in your own teaching. So I'm sure you're all from different areas within initial teacher education at your universities. So you can choose any topic that you like. that is relevant for your area, for what you teach. And then you think about one activity that you might design to embed algorithmic literacy. So it might be in relation to the use of generative AI as one example of an algorithmic system, Or it could be an example of a different algorithmic system, whether it be a Google search engine, a social media feed, et cetera.

[2:12:05] Participant: Thanks.

[2:12:08] Kelly Ilich: You then, we're then going to share ideas with the whole group, and then we're going to complete a SWOT analysis, and I'll explain that a little bit later. So, here's some examples that that you that might get you thinking. So, try to come up with your own. where possible. But if

you can't think of anything, then you can start with these ideas. So for instance, let's say you were teaching a session to your PPG students and one of the activities in that session required them to go online and find some information about something. thing. So that would be a good opportunity to talk about the algorithms that operate within search engines and how it ranks the results. So you could do something where they may, oops, sorry, we've got somebody with a mic on there, just make sure it's turned off.

[2:12:58] Participant: Dad. Whatever. Manasi.

[2:13:09] Kelly Ilich: So you could do an activity where you're getting the search engine to show two different, that shows two different results for two different people. You could get students to cheque an AI summary against its source text. So you know how when you do a Google search now, you get the AI summary right at the top. So you could design an activity where the students need to verify that AI summary. You could ask your students how ranking, rating or prediction was calculated depending on the algorithmic system you're talking about. Or you could cheque an AI worked solution in numeracy and find an error or an unexplained step. So there are just some ideas. You don't have to use those. You can come up with your own. All right, but I'd like you to be able to tell us about the topic that you're teaching and then the activity that you would use to embed algorithmic literacy to start getting your students to think about, you know, how these algorithms operate and the impacts that they have. Does anyone have any questions? So if we do 10 minutes, okay, for this, and then we'll come back and we'll share ideas with the rest of the group.

[2:14:28] Participant: HUM.

[2:14:28] Kelly Ilich: Thanks, everybody. I'll see you in 10 minutes.

[2:14:29] Participant: Okay, thank you. Okay.

[2:15:05] Participant: HUM. Yeah.

[2:24:31] Kelly Ilich: Okay, that's 10 minutes up. Thanks, everybody. I hope you had sufficient amount of time to think about that. Please feel free to continue posting in the comments. I can see some wonderful responses in there so far. Would anybody like to put their mic on and tell us what the topic was and what activity you think would be useful. Anyone like to share? Or I can give you a minute just to have a bit of a read through the responses in the chat if you like. I'll choose one to highlight for you all. And this one comes from Gita. I hope I'm pronouncing your name right, Gita. And the topic is Teaching Fractions for PPG maths students.

[2:25:52] Participant: Yeah.

[2:25:54] Kelly Ilich: Oops, somebody's got their mic on. Let's just turn that off.

[2:25:54] Participant: Ohh.

[2:26:00] Kelly Ilich: And the activity is that students will have to ask an AI tool to solve and explain a fraction problem. For example, comparing or adding unlike fractions. They then cheque the worked solution step by step, find the first error or unexplained step, Correct it, and then explain how they would teach it to primary students. So that's great. To extend on that, you could also think about asking your PPG students to reflect on why they found that error from that AI tool in terms of perhaps what the AI tool is optimised for or the various inaccuracies that we might need to be mindful of, or the limitations of it rather, when it comes to a subject like mathematics. So that was a wonderful example. Thank you for sharing. Would anybody else like to share?

[2:27:03] Participant: Ohh, I'll try Kelly.

[2:27:05] Kelly Ilich: Thank you.

[2:27:06] Participant: Okay, based on my experience last semester, I teach the principles of assessment. And then I choose a topic related to the principle is that the principle of assessment using culturally responsive teaching or

[2:27:14] Kelly Ilich: Ah, good.

[2:27:25] Participant: CRT. And then I will ask my student to use AI to evaluate the responses or the result of AI about certain

[2:27:25] Kelly Ilich: Thank you. HUM.

[2:27:45] Participant: English task, yeah, or certain English, yeah, task, and then they need to evaluate the responses of the AI, yeah, the responses of AI, and then using the strategies you have already mentioned before.

[2:27:51] Kelly Ilich: Right. HUM. And. Yeah.

[2:28:05] Participant: The strategies to what to to get the the the best responses of the AI.

[2:28:11] Kelly Ilich: Mm.

[2:28:12] Participant: And then, yeah, and after that, I will ask my student to evaluate whether AI assess students' ability fairly, and then...

[2:28:13] Kelly Ilich: Too thick.

[2:28:25] Participant: Sims.

[2:28:31] Participant: Can AI assess scoring related to culture and also linguistics? Yeah, and then should they accept the always accept this course generated by AI and so on and so forth? Yeah, and then my student will be asked also.

[2:28:40] Kelly Ilich: HUM.

[2:28:52] Participant: to analyse using SWOT analysis, yeah, about the strength from the result of AI, and then the weakness, yeah, that the weakness of the AI is really depending on the prompt and also the rubrics that they uploaded into AI.

[2:28:55] Kelly Ilich: Yes. HUM. HUM. HUM.

[2:29:15] Participant: AI tool. And then the opportunities is that the result of AI enhance, not enhance, but make them possible to modify and design more inclusive assessment practises.

[2:29:28] Kelly Ilich: Annabel.

[2:29:32] Participant: And the last one, if... about the threat, yeah, if they trust AI scores too much, they will misunderstand the cultural expression and also the language variation resulted from the result of AI. I think that's my experience.

[2:29:46] Kelly Ilich: I think. Amazing. Fantastic. Sounds like you're doing some really great work in that area. Thank you so much for sharing your expertise. And that reminds me because, yes, I teach a similar unit about principles of assessment and assessment practises and that kind of thing. And it's a great opportunity to bring in algorithmic literacy. even just through something as simple as a discussion-based activity where you're just asking a question like, would you use generative AI to mark your students' work and to provide feedback? Why or why not? And then that brings in that discussion of the various capabilities of it and the various limitations. of the algorithms that operate

behind the generative AI. Fantastic, thank you. And Agostina has got a great example in the chat here as well. So I've just lost it. Give me a sec. If you're online, [name], and you want to put your mic on and explain, feel free, because I've just it's just disappeared in my chat. Otherwise, if anybody else wants to jump in and share an example. I found it. Designing and implementing formative assessment in elementary schools. And so here, Agostina has said that they ask the PPG students to investigate how AI algorithms interpret students' learning data. So this is a really great

[2:31:38] Participant: HUM.

[2:31:43] Kelly Ilich: example of an activity that you can do with your PPG students anytime you're talking about interpreting student data, this is a great opportunity to bring in algorithmic literacy. So how have you decided to do that? It is by asking students to investigate how AI interprets students' assessment data and generates learning recommendations. Students then compare AI generated recommendations with their own professional judgments, identifying possible biases and evaluating whether the recommendations are appropriate for different learners. So again, I think that is a really terrific example of how we can be embedding algorithmic literacy into the sessions that we're already teaching our students. Thank you so much for sharing those, everybody. And again, I'm mindful of time, so I'm going to move on and we might come back to those SWOT analyses. So in terms of what my research showed in regards to Australian teachers, as I've already mentioned, they knew a lot about algorithmic systems and their impacts. They cared about this as a topic and they thought it was really important to teach it to their students, but they found how to teach it the hardest thing. All right. And so you might recognise this Venn diagram. So this is the TPAC framework, technological, pedagogical and content knowledge. By Koul et al. So, in terms of applying this framework to the research. The content knowledge was good, the technological knowledge was good, but it was the pedagogical knowledge that they really struggled with. They weren't sure what was the best way of teaching algorithmic literacy, what kind of strategies and approaches would be appropriate, especially for younger year groups.

[2:33:40] Participant: HUM.

[2:33:50] Kelly Ilich: And, you know, they were also worried about things like access to technology and things like that where they may be in schools where that was limited. So in terms of what they wanted to see is that some teachers asked to see it modelled. Many wanted ongoing learning, not one-off workshops. The research does show us that algorithmic literacy education works best when it is woven into existing subjects, not treated as a separate topic. So all of those examples that you've just provided us are excellent examples of how it can be integrated into the sessions that we're already teaching across our programmes. Okay, so we don't need specialised topics or specialised units necessarily to teach this. It can be integrated across all of the units that we're teaching. And that's the same thing that happens in the school setting as well, whether it be primary or secondary. The research shows that it's best understood when it is done within this context of the topics that we're already teaching. And Finland is a great example of how they're doing this. They're treating this as a cross-curricular priority, where they're integrating algorithmic literacy as a part of information and media literacy across all of the subjects that they teach. And as a result, they are performing, then the number one performing nation in Europe when it comes to being able to identify misinformation online. The other thing that the research showed is that a lot of algorithmic literacy education does not actually need to have access to technology or devices. And this is especially the case with younger year groups, because you can teach this through telling stories,

through playing games, a lot of unplugged activities, a lot of discussion-based activities. work really well for teaching this. So they don't necessarily have to be using technology to be able to be taught about algorithmic literacy. But initial teacher education is really critical here for preparing teachers to go out and teach this well. The other thing that was found in my research is that agency is a really critical component of teachers feeling capable to teach this. So if the agency is constrained in any way, then that is going to impact how capable they feel to teach algorithmic literacy. Now, there's six modalities of agency, according to the research by Yakama, as cited in Nagel. Now, these six modalities are to know, to want, to feel it matters, to see the possibility, to have to, and to be able to. Now in Australia, the research showed that teachers know about algorithmic literacy. They want to teach it. They feel that it matters, that it's really important. They can also see the possibilities because they were able to tell me where it would fit in to the subjects that they already teach. but they were constrained in two important ways. They didn't have to. Okay, so in our curriculum, there is no clear requirement to teach algorithmic literacy explicitly. We have a digital literacy general capability. But there's very little mention of algorithms within our curriculum. So because they don't have to, it's not prioritised. There's all these other competing priorities that they have to address first that is deemed to be more important than providing algorithmic literacy education. And then to be able to was also impacted because of all of the external constraints that they're operating within. So I'm not sure how it is in Indonesia, but in Australia, teachers find it really hard to find time. Okay, very, very busy. They're trying to fit in a lot every single day. We've got a very crowded curriculum. So time is a constraint. Resources, they don't have access to good resources for teaching algorithmic literacy. In some areas, technology is very limited. Sometimes they don't get the support to teach this from their principal or colleagues. So to be able to was also a big constraint here. So when you're considering your context in Indonesia and how capable you think your PPG students will be to teach algorithmic literacy when they go out into their own classrooms,

[2:38:39] Participant: Okay. The. Yes. HUM.

[2:38:55] Kelly Ilich: try to consider it through this agency lens. So what modalities would be fulfilled in that context and what modalities may be at risk within your context. Okay, does anyone have any questions at this stage? All right. So in any of the strategies that you use to embed algorithmic literacy in your sessions, these are some different things that you could consider doing. So we've already talked a little bit about teacher modelling, so thinking aloud, thinking through the decisions that you're using when it comes to your use of generative AI. in front of your students. So talking through to them why you're using the tool, what you're asking of the tool, what you are going to cheque before you use the answer, what you changed, what you rejected and why, and then what you chose not to share with it. All of these are really important things to be talking through with your students and modelling best practise. Okay, and through that, they're going to be able to pick up and emulate those best practises for their own teaching.

[2:40:13] Participant: HUM.

[2:40:19] Kelly Ilich: Metacognitive strategies, so thinking about your thinking. So prompts for doing this is before you use generative AI, think about what are you expecting to get from it? What do you already know? During, what is this tool optimised for? Okay, we've spoken a lot about that this session. And is it agreeing too easily? Okay, and after, what did I change? Or what would change my mind? All right. And then when you're embedding algorithmic literacy, when you're designing your activities, you can think about using some of these design moves. So you could put a question to

students that they must ask of an algorithmic output. So for instance, if you're asking your students to use AI in some way in your classroom, get them to think of a question that they could use to probe or to challenge its output. a cheque that they must perform. So maybe you're going to ask them to compare the output with another tool or to compare the output that they got compared to the person sitting next to them. Maybe as part of your activity, you're going to get them to verify the information that the AI provided them. Number 3, you might get them to identify an ethical choice that they need to justify. All right, so again, this is going to depend on the type of activity that you're doing, but try to get them to think about some of the ethical implications that come along with using these AI tools and then justify why they're using it. And then finally, process evidence.

[2:42:08] Participant: HUM.

[2:42:10] Kelly Ilich: they must submit. So get them to submit the prompts, the replies, and then what they changed, you know, what they might have edited from the reply or how they might have re-prompted the tool to get a better response. When you're assessing how well your students are doing in terms of developing their AI literacy, this is a simple guide that you can use to determine their progress. All right. So in terms of the cognitive dimension, we're really looking for them to get to this final column where they're demonstrating it.

[2:42:35] Participant: I. ED. Yeah, HUM.

[2:42:50] Kelly Ilich: Okay, so can they explain how a specific algorithm works, who built it, and what it's optimised for? For the effective dimension, can they ask who benefits here? What's hidden? Is this agreeing to easily and be able to justify their concerns? And for the behavioural dimension, are they actively adapting their behaviour? And can they explain the trade-off?

[2:43:02] Participant: Hold on, please. That's right.

[2:43:13] Kelly Ilich: that they made. All right, and can they also talk to other people about how they're using AI and advocate for better usage of it or more ethical or more responsible usage of it? All right. So what we're going to move on to now is this final activity. Now this is going to lead towards your final assessment. All right. So this activity is going to bring everything together and help you to summarise what you've just learnt, and this will then feed into your final assessment and help you to prepare for that. Okay, so in your individual writing, you will need to identify the session or the topic that you've decided to focus on. The system that you'll use, so let's keep it to generative AI here so you can identify, you know, which generative AI platform you're choosing to use, what it's optimised for, what data it collects, the strategy you will use to challenge the algorithm, The embedded teaching strategy, and then how you will monitor its impact. All right, so that's going to be the last 20 minutes of our session today, although I might come back online about 5 minutes before 6. All right, and that will be an opportunity to ask any final questions before we finish for today. Alright, I'll leave you to your individual writing. Thank you, everybody.

[2:45:01] Participant: Yeah.

[2:58:35] Participant: Yeah.

[2:59:26] Kelly Ilich: Okay, thank you everybody. Hopefully you've had some time to make a good start on that. If you haven't quite finished, that's okay. You might want to spend a little bit of time after this session finishes to just complete that. But I thought I'd just spend the last five minutes wrapping up and then providing some opportunities for any final questions. So, you know, we've talked a lot about algorithms in this session in terms of what an algorithmic system is optimised for.

So, you know, we talked about, for instance, with social media being based on that attention model. Whereas the generative AI is more based on that attachment model. OK, trying to make you depend on it through being really helpful. OK, we need to ensure that we're treating any praise that comes back from the... generative AI as a prompt to test it further. Okay, it provides us with some clues that perhaps the chat bot is prioritising being helpful rather than providing us with balanced and accurate. Information. It's great that we've got really strong scores in terms of our cognitive dimensions and our affective dimensions, but what we really need to focus on is translating that into our action. All right, so knowing matters most when it changes what you do. So as I explained in terms of my Australian study, this is definitely the case. And it would be really interesting to know if that's also the case in your context in terms of what you're observing with your students. So focusing on the actions to try and help your students navigate these systems is the most critical part of the work. And one of the ways we can do this is through modelling our own AI use. So talking through the decisions that we've made, the thinking behind the use of AI with our students really does help to model best practise. And of course, as you already know, AI can be a really useful scaffold and tool for the teacher, but it may not be the best judge of our students' work. So, professional judgement is always key. Okay, does anyone have any questions at all before we do finish up for today's session? You've got your hand up there.

[3:02:17] Participant: Okay, thank you, Kelly. I have one question about data privacy. When we type our own information into AI, for example, while drafting a research proposal or uploading a lesson plan, does that information go into the company database?

[3:02:19] Kelly Ilich: No problems, yes. Uh-huh. HUM.

[3:02:38] Participant: It is possible if, I mean, another user to recall our information or ideas by asking with similar keywords. Thank you.

[3:02:51] Kelly Ilich: Okay, so I think I understand your query. So yeah, all of the information that you put into these models, the web-based models, the open access models, all of that information is collected by the companies who control these models. And that is used to train AI And in the case of advertising, it can be used to help with targeted advertising and things like that. Yeah. Does that answer your question?

[3:03:23] Participant: Yes, thank you, Kelly.

[3:03:23] Kelly Ilich: Yep. No problems. Anything else?

[3:03:28] Participant: Um, me, Kelly.

[3:03:31] Kelly Ilich: Yes.

[3:03:31] Participant: Okay, I'm not going to ask you a question, but this is my reflection about what you have explained in this session today. So algorithm literacy is very, very, very important for us as human being, not only to

[3:03:32] Kelly Ilich: Is Jane. Thanks.

[3:03:50] Participant: as teachers, but also, yeah, the most importantly to us as teachers, yes, in Indonesia, because your explanation today makes me flash back to what happened, certain cases happen in other parts of the world. I remember that in what, in certain areas in the world like in the US, yeah, that a family, what take the case of certain AI responses to the court just because the AI gave the suggestion of mixing certain chemical substances and the ChatGPT say,

[3:04:14] Kelly Ilich: And then? HUM. Yeah.

[3:04:29] Participant: it's okay to drink it. And then without testing the understanding and acting and testing the responses, the teenagers drank the chemical response and he died. And in the other parts of the world also like that. So it is very, very important. So I think

[3:04:32] Kelly Ilich: Mmh. Yeah. Yeah. Yep.

[3:04:50] Participant: To you, yeah, thanks so much for reminding us how important this principle is.

[3:04:56] Kelly Ilich: Thank you so much, Izzy. I really appreciate your reflection and your participation today. Fantastic.

[3:04:58] Participant: HUM.

[3:05:05] Kelly Ilich: Okay, I think that brings us to 6 o'clock on the dot. So thank you everybody for joining and for participating in this session. It was a wonderful experience for me being able to chat with you all and gain the benefit of all of your knowledge and amazing ideas as well, because you're obviously coming into this session with you know, really strong background knowledge about this topic. And it's always interesting to be able to discuss how we would go about teaching it in our own sessions as well. So it's helped me to reflect and think more about this myself too. All right. Was there anything that we needed to do at the end in terms of assessment, sorry, attendance? Craig did Mention something about that to me. No. Okay. No problems then. All right. Well, thank you again, everybody, and I hope you enjoy the rest of the programme. I'll be coming back with Craig to do another session later on down the track. So I look forward to seeing you all again then.

[3:06:27] Participant: Thank you, Kelly.

[3:06:27] Kelly Ilich: All right.

[3:06:28] Participant: Thank you, Kelly. Thank you.

[3:06:29] Participant: Thank you, everyone. Thank you, Kelly. Thank you so much. Thank you. Thank you.

[3:06:29] Kelly Ilich: Thanks, everyone. Thank you, Kelly.

[3:06:30] Participant: Thank you, Kelly. Thank you, Kelly.

[3:06:31] Participant: Thank you, Kelly. Thank you, Kelly. Thank you, Kelly. Thanks.

[3:06:35] Participant: Thank you, Kelly.

[3:06:36] Participant: Thank you, Karen.

[3:06:37] Kelly Ilich: Okay.

[3:06:38] Participant: Thank you, Kelly. See you.

[3:06:39] Participant: Is there any for the day?

[3:06:39] Kelly Ilich: Play.

[3:06:41] Participant: See you.

[3:06:41] Kelly Ilich: Bye.

[3:06:50] Participant: No photo session.